

BEN WILLCOX

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PROFESSIONAL SUMMARY

Data analyst and analytics product builder with 4+ years owning production SQL, R, and Python workflows across revenue cycle, physician productivity, scheduling, and clinical operations. Supports approximately 100 administrators and 200+ physicians through 6 dashboards, 20+ reports, and 20+ automated workflows, while building deployed full-stack analytics and applied modeling projects.

TECHNICAL SKILLS

Analytics & Data: SQL, Python (Pandas, NumPy, scikit-learn), R (Shiny, R Markdown), PostgreSQL, data modeling, statistical modeling, data quality validation

Applications & Infrastructure: Power BI, Grafana, Next.js, TypeScript, React, FastAPI, Supabase, Drizzle ORM, Streamlit, Recharts, Plotly, Git

Applied Modeling: regression, Elo systems, Monte Carlo simulation, clustering, model calibration, computer vision evaluation

EXPERIENCE

STARS Advanced Radiology Services

Grand Rapids, MI (Remote)

Data Analyst

Apr 2022 - Present

- Own an analytics portfolio across billing, physician productivity, scheduling, and clinical operations, supporting approximately 100 administrators and 200+ physicians through 6 dashboards, 20+ reports, and 20+ automated workflows.
- Built and maintain a multi-tab R Shiny billing application end to end, from SQL transformations and KPI design through training and enhancements; used daily by 25-30 employees and executives as Days A/R fell from above 60 to 29-32.
- Replaced a manual query-to-Excel workflow with an interactive Critical Results dashboard that prioritizes urgent physician support requests; compliance improved from approximately 89% to above 98%.
- Contributed to a Python regression model that evaluates modality, RVU workload, and RVUs per FTE every 2 minutes, enabling surge warnings and data-driven on-call staffing.

Byrne Electrical Specialists, Inc.

Rockford, MI

Data Science Intern - Three Summer Terms

Summers 2019-2021

- Built PostgreSQL-backed IoT sensor pipelines, Grafana and Power BI dashboards, and SPC alert logic for robotic assembly OEE, contributing to a 15% improvement in machine OEE.
- Developed an SPC model for crimp-terminal measurements and partnered with developers to embed alerts in the machine HMI, enabling preventative adjustments before out-of-spec production created scrap or electrical safety risks.

SELECTED PROJECTS

ValoMapped - Esports Elo Analytics Platform | *Next.js, TypeScript, PostgreSQL, Drizzle ORM, Recharts*

- Built and deployed a VCT analytics platform with map-specific Elo ratings, match predictions, player ratings, and Monte Carlo tournament simulations; published a chronological backtest across 3,818 maps with validation/test separation, calibration, Brier score, log loss, and confidence intervals.

World Cup 2026 Draw Simulator | *TypeScript, React, recursive backtracking, Monte Carlo simulation*

- Built a constraint-tested draw engine with versioned rules, regression fixtures, generated property tests, and seeded million-trial artifacts; exposes group, opponent, and travel probabilities while documenting sampler assumptions and limitations.

HoopStats - Basketball Video Analytics Pipeline | *Python, PyTorch, OpenCV, Roboflow, SAM2*

- Built detection, tracking, team classification, court homography, shot-event export, ground-truth annotation, and evaluation workflows for broadcast basketball footage, with unit-tested event and scoring logic.

Voting Paradigm - Voting Systems and Social Choice Demos | *Python, FastAPI, Next.js, TypeScript, Supabase*

- Built a hosted portfolio comparing voting methods, apportionment, districting, and preference-agent replay, backed by a tested Python package covering plurality, approval, IRV, Borda, Ranked Pairs, score, and quadratic voting.

EDUCATION

Michigan State University

Dec 2021

B.S. Economics | Minor in Quantitative Data Analytics